**MILESTONE 1**
**CLIENT DESIGN REVIEW**

CLI-027 — MOTSHELO HIGH SCHOOL (TAUNG)
Computer Networking Project — CMPG 325

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|---|---|
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| Project ID | CMPG325-2026-027 |
| Client ID | CLI-027 |
| Organisation | Motshelo High School(Taung) |
| Industry | Education |
| Assigned Address Block | 172.30.8.0/23 |

## 1. Client Requirements

This design is specifically for Motshelo High School (Taung), Client CLI-027. The solution uses the assigned 172.30.8.0/23 network and is designed for implementation and simulation in Cisco Packet Tracer.

### 1.1 Core client requirements

- Provide appropriate network connectivity and services for the school's assigned scenario.
- Use 172.30.8.0/23 as the basis of the IP addressing plan.
- Design an appropriate topology and device arrangement.
- Configure routers, switches, end devices, and other required nodes.
- Demonstrate successful connectivity and testing in Cisco Packet Tracer.
- Document significant design decisions and evidence in GitHub.

### 1.2 Assigned network challenge

Basic OSPF (single-area dynamic routing). The school routing domain will use OSPF Area 0 between the Layer-3 core and edge routers. OSPF will advertise the internal VLAN networks and the required routed transit links. Internet default routes will be handled separately so that the primary and backup WAN paths can be controlled and verified.

### 1.3 Design constraint

A backup Internet path is required for the Finance function. R1/ISP1 is the preferred primary path and R2/ISP2 is the backup path. The design uses tracked/default-route logic (or equivalent Packet Tracer-supported floating/default routing) so that traffic can move to the backup path when the primary WAN path fails.

### 1.4 Client Change Request – CR13

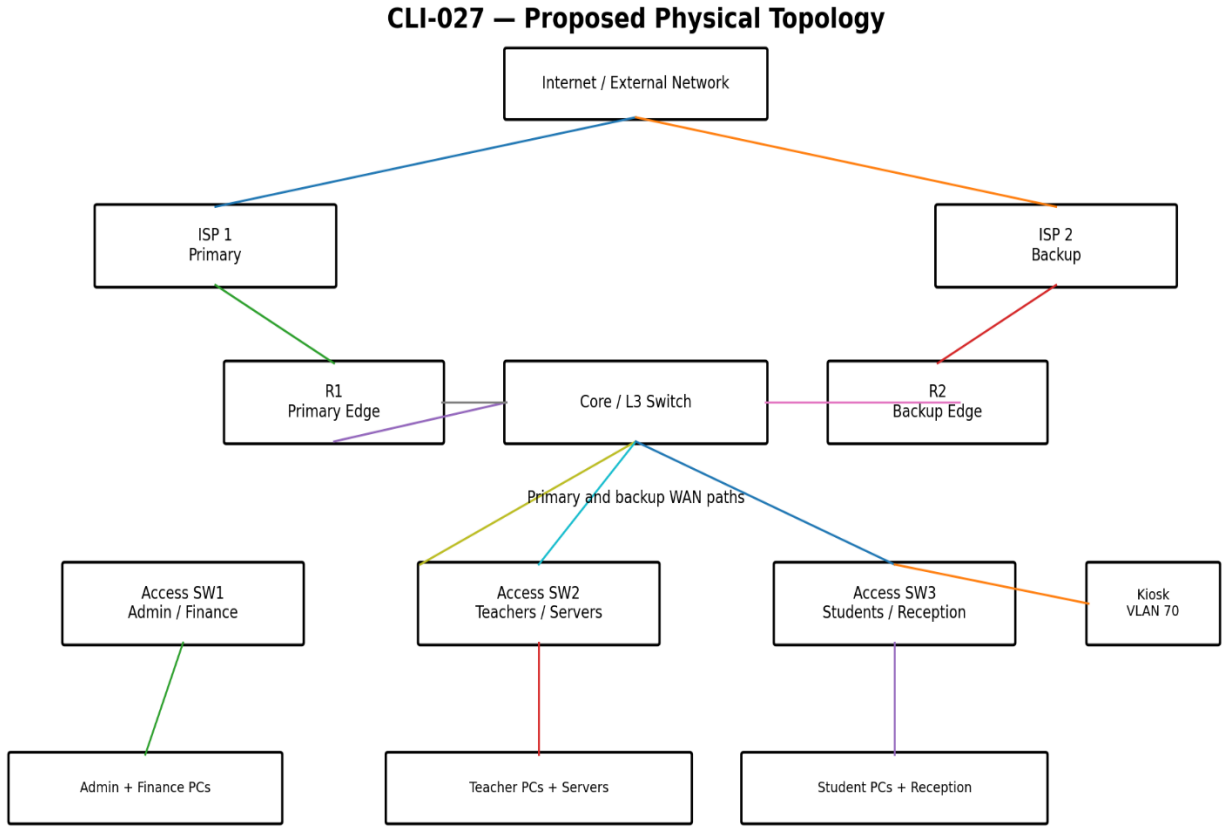
A public self-service kiosk is installed at Reception with Internet-only access. The kiosk is therefore placed in a dedicated VLAN and must be prevented from reaching the school’s internal networks while retaining Internet access.

### 1.5 Testing and Evidence Requirements

- Test relevant end-to-end connectivity.
- Verify the assigned networking challenge (OSPF).
- Capture clear evidence of successful operation/configuration.
- Document important troubleshooting performed.
- Ensure the final Packet Tracer .pkt file opens and reproduces the working solution.
- Maintain an individual GitHub portfolio with a professional README and project evidence.

## 2. Physical Topology

The physical topology shows the proposed network devices and their connections. Two simulated Internet providers are used. The Layer-3 core provides inter-VLAN routing, while R1 and R2 provide separate primary and backup WAN paths. NAT/PAT is required on the edge router(s) so private school addresses can access the simulated Internet.




### 2.1 Proposed Physical Devices

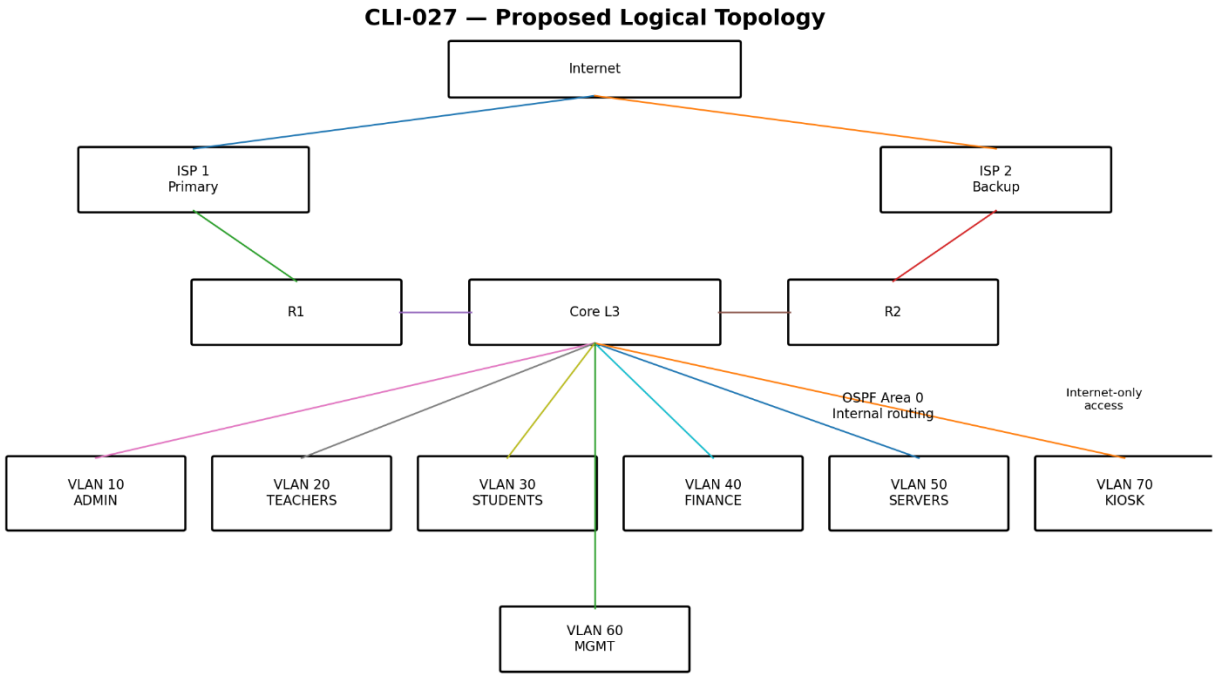
| Device | Quantity | Purpose |
|---|---|---|
| R1 — Primary Edge Router | 1 | Primary Internet/WAN path |
| R2 — Backup Edge Router | 1 | Backup Internet/WAN path |
| Core Layer-3 Switch | 1 | Inter-VLAN routing and core connectivity |
| Access Switches | 3 | Connect departmental/end-user devices |
| ISP Router 1 | 1 | Simulated primary Internet |

| | | |
|---|---|---|
| ISP Router 2 | 1 | Simulated backup Internet |
| Server(s) | 1+ | Internal school services |
| End-user PCs | As required | Administration, teachers, students and Finance |
| Reception Kiosk | 1 | Public self-service Internet-only device |

Implementation note: device quantities can be adjusted to match the final Packet Tracer scenario, but the same logical structure and addressing plan should be maintained.

### 3 Logical Topology

The logical topology separates user groups and services using VLANs. Inter-VLAN routing is performed by the Layer-3 core switch. The Reception kiosk is isolated in its own VLAN and an ACL on the routed interface prevents access to internal school networks while permitting Internet-bound traffic.




### 3.1 VLAN design

| VLAN | Name | Purpose | Security/ Design Intent |
|---|---|---|---|
| 10 | ADMIN | Administration | Internal staff network; controlled inter-VLAN access |

| | | | |
|---|---|---|---|
| 20 | TEACHERS | Teaching staff | Internal staff network; controlled inter-VLAN access |
| 30 | STUDENTS | Students | Student segment; controlled access to services |
| 40 | FINANCE | Finance | Protected business function; primary + backup WAN path |
| 50 | SERVERS | Internal servers | Internal services; access restricted by policy |
| 60 | MANAGEMENT | Network management | Management-only addresses for network devices |
| 70 | KIOSK | Reception public kiosk | Internet-only; ACL blocks internal school networks |

### 3.2 OSPF Design

- Use Basic OSPF in a single area: Area 0.
- Advertise the internal school VLAN networks and routed core-to-edge transit networks through OSPF Area 0. Do not rely on OSPF to advertise the external ISP networks; use controlled default-route propagation toward the school network.
- Verify neighbour relationships using appropriate Cisco show commands.
- Verify learned routes in the routing table.
- Use ping/traceroute tests to demonstrate end-to-end reachability.

### 3.3 Finance Backup Internet Design

Finance is connected to the school routed network and has access to two WAN paths. R1/ISP1 is the preferred primary path and R2/ISP2 provides the backup path. The implementation must demonstrate the primary path in normal operation and failover to the backup path when the primary WAN path is disabled. The routing method must prevent a permanently preferred backup route from overriding the primary path.

### 3.4 Reception Kiosk Design

- Place the kiosk in VLAN 70 (172.30.9.64/27).
- Allow the kiosk to reach the Internet.
- Block access from the kiosk to internal school VLANs.
- Do not place the kiosk in Administration, Teachers, Students, Finance, Servers or Management VLANs.

- Implement the Internet-only restriction with an extended ACL applied to the VLAN 70 SVI in the inbound direction. The ACL must deny traffic from the kiosk subnet to the internal school VLAN subnets, permit Internet-bound traffic, and include an appropriate final rule. NAT/PAT on the edge router must translate the kiosk private address when it exits toward the simulated ISP.

**4. IP Addressing Plan**

Assigned address block: 172.30.8.0/23. A /23 contains 512 total addresses, with 510 usable host addresses. The block spans 172.30.8.0 through 172.30.9.255. The VLAN subnets below consume the required user/service networks, while a reserved portion of the block is kept for routed transit links and future expansion.

| VLAN/Link | Name | Network | Prefix | Subnet Mask | Gateway/Endpoint A | Endpoint B / Usable Hosts |
|---|---|---|---|---|---|---|
| 10 | ADMIN | 172.30.8.0 | /26 | 255.255.255.192 | 172.30.8.1 | 62 hosts |
| 20 | TEACHERS | 172.30.8.64 | /26 | 255.255.255.192 | 172.30.8.65 | 62 hosts |
| 30 | STUDENTS | 172.30.8.128 | /26 | 255.255.255.192 | 172.30.8.129 | 62 hosts |
| 40 | FINANCE | 172.30.8.192 | /26 | 255.255.255.192 | 172.30.8.193 | 62 hosts |
| 50 | SERVERS | 172.30.9.0 | /27 | 255.255.255.224 | 172.30.9.1 | 30 hosts |
| 60 | MANAGEMENT | 172.30.9.32 | /27 | 255.255.255.224 | 172.30.9.33 | 30 hosts |
| 70 | KIOSK | 172.30.9.64 | /27 | 255.255.255.224 | 172.30.9.65 | 30 hosts |
| Transit 1 | CORE-R1 | 172.30.9.96 | /30 | 255.255.255.252 | Core: 172.30.9.97 | R1: 172.30.9.98 |
| Transit 2 | CORE-R2 | 172.30.9.100 | /30 | 255.255.255.252 | Core: 172.30.9.101 | R2: 172.30.9.102 |
| Transit 3 | R1-ISP1 | 172.30.9.104 | /30 | 255.255.255.252 | R1: 172.30.9.105 | ISP1: 172.30.9.106 |
| Transit 4 | R2-ISP2 | 172.30.9.108 | /30 | 255.255.255.252 | R2: 172.30.9.109 | ISP2: 172.30.9.110 |

**4.1 Addressing Conventions**

- Use the first usable address of each subnet as the default gateway.
- Use static addresses for infrastructure devices and servers where appropriate.

- Use DHCP for ordinary end-user PCs if DHCP is implemented in the final Packet Tracer design.
- Reserve the unused portion of 172.30.9.96/27 for routed point-to-point links and future expansion. Use /30 subnets from this reserved range for router-to-router transit links so that the WAN/core routing design is explicitly addressed.
- Keep the same addresses in the Packet Tracer file, topology diagrams and GitHub documentation.

4.2 Example of host addressing

| **Device/Role** | **VLAN** | **Example IP** | **Default Gateway** |
|---|---|---|---|
| Admin PC 1 | 10 | 172.30.8.10 | 172.30.8.1 |
| Teacher PC 1 | 20 | 172.30.8.70 | 172.30.8.65 |
| Student PC 1 | 30 | 172.30.8.140 | 172.30.8.129 |
| Finance PC 1 | 40 | 172.30.8.200 | 172.30.8.193 |
| Internal Server 1 | 50 | 172.30.9.10 | 172.30.9.1 |
| Management Device Example | 60 | 172.30.9.40 | 172.30.9.33 |
| Reception Kiosk | 70 | 172.30.9.70 | 172.30.9.65 |

4.3 **Routing, NAT and ACL corrections**

- OSPF: Configure Area 0 between the Layer-3 core and R1/R2. Advertise VLAN 10–70 networks and the core-to-edge transit links. Verify with show IP OSPF neighbor and show IP route OSPF.
- Default routing: Configure a preferred default route toward ISP1 and a backup/floating default route toward ISP2, or use an equivalent tracked failover design supported by Packet Tracer. Verify that the backup becomes active when the primary WAN path fails.
- NAT/PAT: Configure NAT overload on the active edge router so private VLAN addresses can access the simulated Internet. Include VLAN 70 in the translated inside networks.
- Kiosk ACL: Apply an extended ACL to the VLAN 70 SVI inbound. Deny the kiosk subnet to each internal VLAN subnet (172.30.8.0/24 and 172.30.9.0/26, plus the reserved internal transit addresses as appropriate), then permit Internet-bound traffic. Verify both blocked internal access and successful Internet access.
- Transit addressing: Use the /30 point-to-point subnets listed above for the core-to-edge and edge-to-ISP links. This removes ambiguity about router interfaces and ensures the OSPF topology is fully addressable.
- Testing: Capture evidence for OSPF adjacency, routing tables, normal Finance Internet access, primary-link failure and backup recovery, kiosk-to-internal blocking, and kiosk Internet access.

**5. Initial GitHub Respository**

https://github.com/kearabilekwakwa00-oss/CLI-027-Motshelo-High-School-Network.git